System Flow — Purpose, Library & Example

1️⃣ Camera Input

Purpose: Capture live video stream from Hikvision camera
Library: OpenCV
Example:

- Read RTSP feed
- Convert into frames for processing

```
cap = cv2.VideoCapture("rtsp://user:pass@192.168.1.10:554/Streaming/Channels/101")
```

2️⃣ Face Detection

Purpose: Detect face location in frame
Library: MediaPipe
Example:

- Find bounding box around face
- Return coordinates

```
mp_face_detection.process(frame)
```

Output:

```
[x, y, width, height]
```

3️⃣ Face Alignment

Purpose: Normalize face angle & orientation
Library: InsightFace / OpenCV
Example:

- align eyes horizontally
- crop face properly

This improves recognition accuracy.

4️⃣ Face Recognition Model

Purpose: Identify person
Library: InsightFace (ArcFace model)

ArcFace converts:

Face image → 512 vector embedding

5️⃣ Generate Face Embedding

Purpose: Convert face into numerical representation
Library: InsightFace

Output example:

[0.132, -0.551, 0.221, ...512 values]

This becomes the “face signature”.

6️⃣ Similarity Search

Attendance systems don't “compare images”.

They compare embeddings.

Library: FAISS
Purpose: Fast vector matching

Process:

New embedding → search DB embeddings → closest match

7️⃣ Matched Person ID

Purpose: Identify employee from vector match

Example:

embedding match → employee_id = 104

8️⃣ Attendance Engine

Purpose: Decide:

- present
- late
- already marked
- duplicate detection

Logic:

if first entry → mark present

if after schedule → mark late

9️⃣ Database

Library: PostgreSQL

Tables:

- employees
- embeddings
- attendance_logs

🔟 Django Dashboard

Purpose:

- admin panel
- reports
- live attendance
- search logs

📊 Clean System Flow Table (Documentation Ready)

Step	Component	Purpose	Library
1	Camera RTSP	Capture video stream	OpenCV
2	Face Detection	Locate face in frame	MediaPipe
3	Face Alignment	Normalize face	InsightFace/OpenCV
4	Recognition Model	Identify person	InsightFace ArcFace
5	Embedding Generator	Convert face → vector	InsightFace
6	Similarity Engine	Match embeddings	FAISS
7	Identity Mapping	Get employee ID	FAISS + DB
8	Attendance Engine	Apply attendance logic	Python
9	Database	Store records	PostgreSQL
10	Web Dashboard	Monitoring & reports	Django